

Zoha Rakotomalala

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Software engineer with a mathematics and data background, drawn to machine learning and the science behind intelligent systems. I like hard problems, new methods, and turning ideas into things that work.

Skills

Machine Learning & Data	Classical ML (decision trees, classification) · scikit-learn · pandas · Dataiku (Core Designer) · statistics & data modeling · SQL · Amazon Redshift · ElasticSearch
Programming	Python · C++ · Java · TypeScript · JavaScript · SQL · React Native · Angular · C# · HTML5 · CSS3
Cloud & DevOps	AWS (CDK, Lambda, DynamoDB, S3, CloudWatch) · Docker · GitHub · CI/CD
Languages	French · English · Spanish · Urdu · Hindi

Work Experience

Software Development Engineer · Amazon

XENOS / CROSSMARKET PRICING · SELLER-FACING TEAM

Paris, France | Jan 2026 – Present

- Used **AI coding agents** (Claude, Kiro, internal tooling) with earlier work as context to run backend migrations in parallel, **cutting per-migration effort about 26%**
- Delivered a migration workstream in the 2026 regional-exit program for pricing services (4 backend services moved off a legacy AWS region, one legacy data-warehouse cluster deprecated), **cutting the region footprint 82%, one month ahead of target**
- Re-platformed fee computation onto a new **centralized fee engine** with reusable cross-region networking constructs

PRICE PERCEPTION · CUSTOMER-FACING TEAM

Amsterdam, Netherlands | May 2024 – Dec 2025

- Designed architecture for a new price-history experience, evaluating tradeoffs between **real-time computation and pre-computed approaches** for latency
- Launched feature experiments using **A/B testing** to measure customer impact; authored a **Correction of Error**, driving root-cause analysis across systems
- Presented at an internal conference on **Model Context Protocol** and its implications for AI systems; supervised a software engineering intern for 5 months

Software Engineering Intern · Dataiku [video demo](#)

Paris, France | Jul 2023 – Dec 2023

- Built a 1900-line feature for a data-science platform, generating foreign-key relation graphs across tables: indexed keys with JDBC into ElasticSearch, generated the graph with Graphviz, and built an Angular component to explore it interactively

Data Analyst Intern · Dashlane / Spark.do

Paris, France | Jul 2021 – Jun 2022

- Built a **predictive model for user retention** to guide the sales team; ran data analysis, modeling, and visualization with Tableau, SQL Server, and Amazon Redshift
- Automated weekly performance reports with the Metabase and Airtable APIs, **cutting preparation time 50%**

Selected Projects

Loan-Default Prediction — Machine Learning (Python)

2023

- Trained a **decision-tree classifier on ~208K loan records**; feature engineering, validation, and threshold tuning **lifted recall from 0.29 to 0.55**. Dataiku Core Designer certified

Protein-to-DNA Sequence Transcriber (Java)

Dec 2022

Genetic Data Automation course

- Probabilistic translation from amino-acid sequences to codons, with edge-case handling and validation against biological-rule checkers

Palette — Cross-Platform Mobile App (React Native / TypeScript)

2025 – Present

Personal side project · [repository](#) · [site](#)

- iOS and Android app searching **14 museum APIs**, offline-first with cloud sync (Supabase, Zustand, MMKV, conflict resolution); full test suite in CI on GitHub Actions

2D Procedural World Generation (Java)

Dec 2022

Data Structures course · [demo](#)

- Procedurally generated maps with rooms, hallways, and varied terrain; session serialization and mouse/keyboard interaction

Education

Lycée Saint-Louis — CPGE MPSI / MP

2017 – 2019

INTENSIVE TWO-YEAR PREPARATORY CLASSES FOR THE COMPETITIVE ENTRY EXAMS TO TOP FRENCH ENGINEERING SCHOOLS (GRANDES ÉCOLES)

- Advanced **mathematics, physics, chemistry, and computer science** (MPSI then MP) — the applied-mathematics grounding behind my technical work

Triple Degree — EDHEC | SKK GSB | UC Berkeley

Sep 2019 – 2023

Global Economic Transformation & Technology (GETT)

- Haas School of Business, University of California**, Berkeley, USA — Diploma in Global Economic Transformation and Technology (Aug 2022 – May 2023)
- SKK GSB, Sungkyunkwan University**, Seoul, Republic of Korea — Master in Management Studies (Jan – May 2021)
- EDHEC Business School**, Paris, France — Master in Management, Full English track (Sep – Dec 2020)

Relevant courses: AI & ML for Business, Data Science Applied to Finance, Data Structures, Data Analytics, Genetic Data Automation, Introduction to C++ and Python, Web Development

Interests

Technology	Discovering new techniques, especially in AI and machine learning; coding and learning for fun
Reading	Across genres: fantasy, science fiction, dystopias, non-fiction, and sociology
Art	Museums, art history, and paintings
Video Games	RPGs and strategy games